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MATLAB 5

Exercise

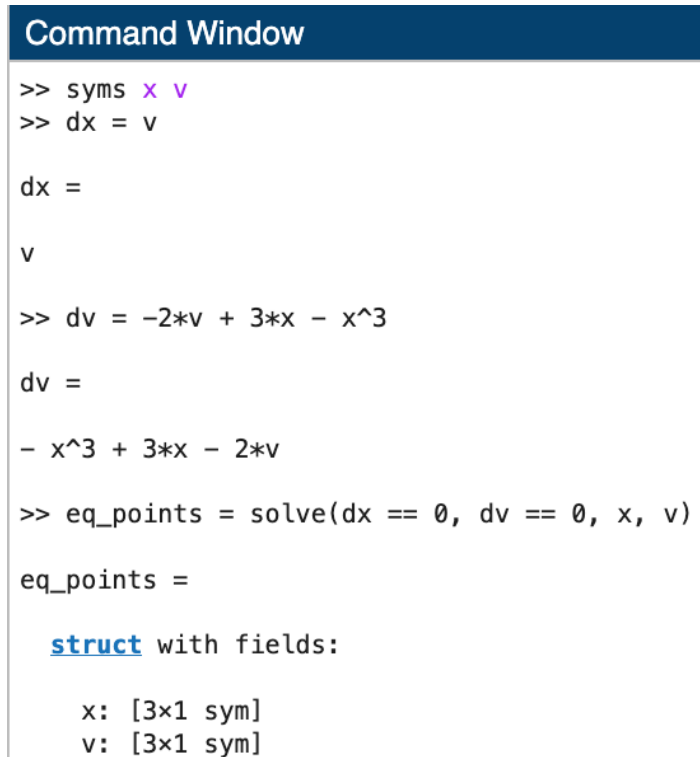
Convert the second-order differential equation $d^2x/dt^2 + 2 dx/dt - 3x + x^3 = 0$ into a first-order system in terms of x and v , where $v = dx/dt$

% We can rewrite this equation: $x'' + 2x' - 3x + x^3 = 0$

% Solving for the highest derivative: $x' = -2x' + 3x - x^3$.

a. Find all equilibrium points.

Command Window Screenshot:

A screenshot of the MATLAB Command Window. The title bar is dark blue with the text "Command Window" in white. The background is white. The text in the window is as follows:

```
>> syms x v
>> dx = v

dx =

v

>> dv = -2*v + 3*x - x^3

dv =

- x^3 + 3*x - 2*v

>> eq_points = solve(dx == 0, dv == 0, x, v)

eq_points =

    struct with fields:

        x: [3x1 sym]
        v: [3x1 sym]
```

b. Use MATLAB to sketch the associated direction field.

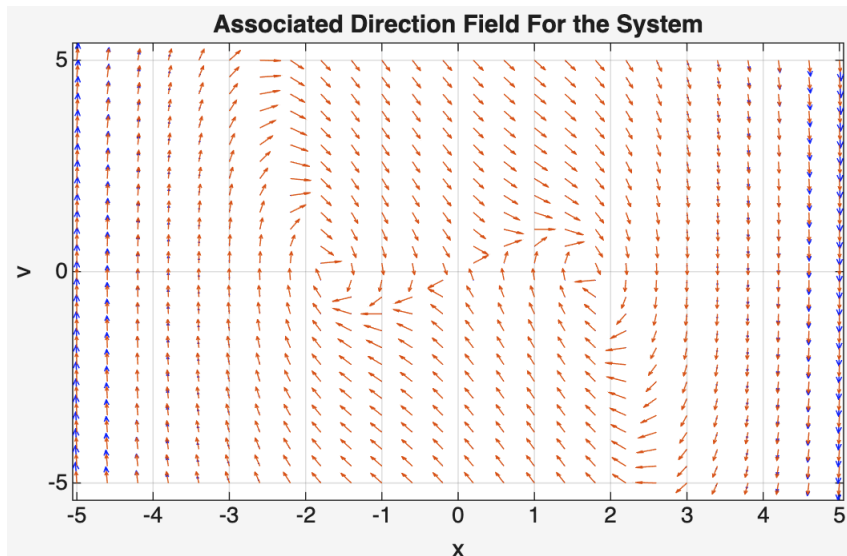
Hint: use `meshgrid(-5:0.4:5,-5:0.4:5)` and `axis tight`

Command Window Screenshot:

Command Window

```
>> [x, v] = meshgrid(-5:0.4:5, -5:0.4:5);  
>> dx = v;  
>> dv = -2*v + 3*x - x.^3;  
>> figure  
>> quiver(x, v, dx, dv, 'b')  
>> axis tight  
>> grid on  
>> hold on  
>> xlabel('x')  
>> ylabel('v')  
>> title('Direction Field for the System')  
>> L = sqrt(dx.^2 + dv.^2);  
>> quiver(x, v, dx./L, dv./L, 0.5)  
>> title('Associated Direction Field For the System')
```

Direction Field Screenshot:



% I included the orange arrows to increase the visibility of the direction field.

c. The following script plots several solution curves over the interval $-3 \leq t \leq 3$

```
f=@(t,P) [P(2); 3*P(1)-P(1)^3-2*P(2)];
a=-2:0.2:2;
b=-2:0.2:2;
for k=1:length(a)
    [t,Pap]=ode45(f, [0,3],[a(k),b(k)]);
    plot(Pap(:,1),Pap(:,2),'linewidth',2)
    hold on
    [t,Pap]=ode45(f, [0,-3],[a(k),b(k)]);
    plot(Pap(:,1),Pap(:,2),'linewidth',2)
end
hold off
axis([-5,5,-5,5])
```

What are the initial conditions?

% The initial conditions in this script are $x(0) = v(0) = a(k) = b(k) \in (-2, -1.8, -1.6, -1.4, -1.2, -1.0, -0.8, -0.6, -0.4, -0.2, 0, 0.2, 0.4, 0.6, 0.8, 1, 1.2, 1.4, 1.6, 1.8, 2)$. There is a for loop that creates solution curves with initial conditions starting at -2 and incrementing forwards and backwards in time.

d. Use the above script to plot a phase portrait together with the direction field of part c.

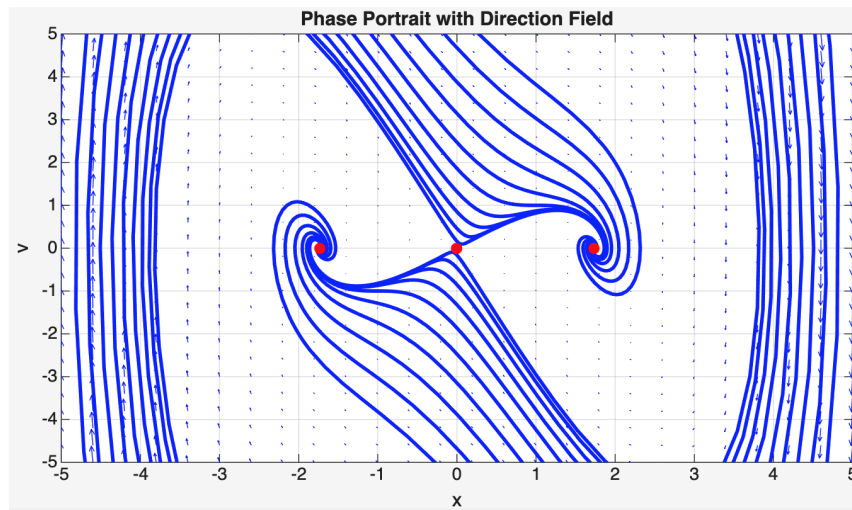
Command Window Screenshot:

Command Window

```
>> f = @(t,P) [P(2); 3*P(1) - P(1)^3 - 2*P(2)];
>> [x, v] = meshgrid(-5:0.4:5, -5:0.4:5);
>> dx = v;
>> dv = -2*v + 3*x - x.^3;
>> figure
>> quiver(x, v, dx, dv, 'b')
>> hold on
>> axis tight
>> xlabel('x')
>> ylabel('v')
>> title('Phase Portrait with Direction Field')
>> grid on
>> a = -2:0.2:2;
>> b = a;
>> for k = 1:length(a)

[~,Pap] = ode45(f, [0, 3], [a(k), b(k)]);
plot(Pap(:,1), Pap(:,2), 'b', 'LineWidth', 2)
[~,Pap] = ode45(f, [0, -3], [a(k), b(k)]);
plot(Pap(:,1), Pap(:,2), 'b', 'LineWidth', 2)
end
>> eq = [0 0; sqrt(3) 0; -sqrt(3) 0];
>> plot(eq(:,1), eq(:,2), 'ro', 'MarkerFaceColor','r', 'MarkerSize', 6)
>> hold off
>> axis([-5 5 -5 5])
```

Phase Portrait Plot:



e. Describe the behavior of the solutions.

% The system has three equilibrium points: a saddle point at $(0, 0)$ and two spirals centered at $(\sqrt{3}, 0)$ and $(-\sqrt{3}, 0)$. The spirals are stable as the curve centers in at the equilibrium points and the saddle is unstable as the curves move away from it. Initial conditions along the diagonal ($a(k) = b(k) = x(0) = v(0)$) show that solutions with positive initial values move toward $(\sqrt{3}, 0)$. and solutions with negative initial values move toward $(-\sqrt{3}, 0)$. Overall, the phase portrait shows that all curves eventually approach one of the two stable spiral equilibria, except those starting on the saddle's stable manifold.